

Internship Report

M1 International Track in France - Life Science & Health

2025/2026

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*Design and construction of a Golden Gate-compatible modular toolkit
for promoter circuit assembly*

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List of Abbreviations

AV Acceptor Vector

CmR Chloramphenicol Resistance

CDS DNA Coding Sequence

CPEC Circular Polymerase Extension Cloning

E. coli *Escherichia coli* K-12

GFP Green Fluorescent Protein

GGA Golden Gate Assembly

KanR Kanamycin Resistance

LVA Leucine-Valine-Alanine (degradation peptide tag)

MEFL Molecules of Equivalent Fluorescein

MoClo Modular Cloning

OD Optical Density

ORI Origin of Replication

PR Right Promoter

PL Left Promoter

P_x Promoter named X

RBS Ribosome Binding Site

SmR Spectinomycin Resistance

SBOL Synthetic Biology Open Language

sfGFP Superfolder Green Fluorescent Protein

TmpR Trimethoprim Resistance

TU Transcriptional Unit (Promoter::RBS::CDS::Terminator)

TF Transcription Factor

WPS Whole-Plasmid Sequencing (Oxford Nanopore)

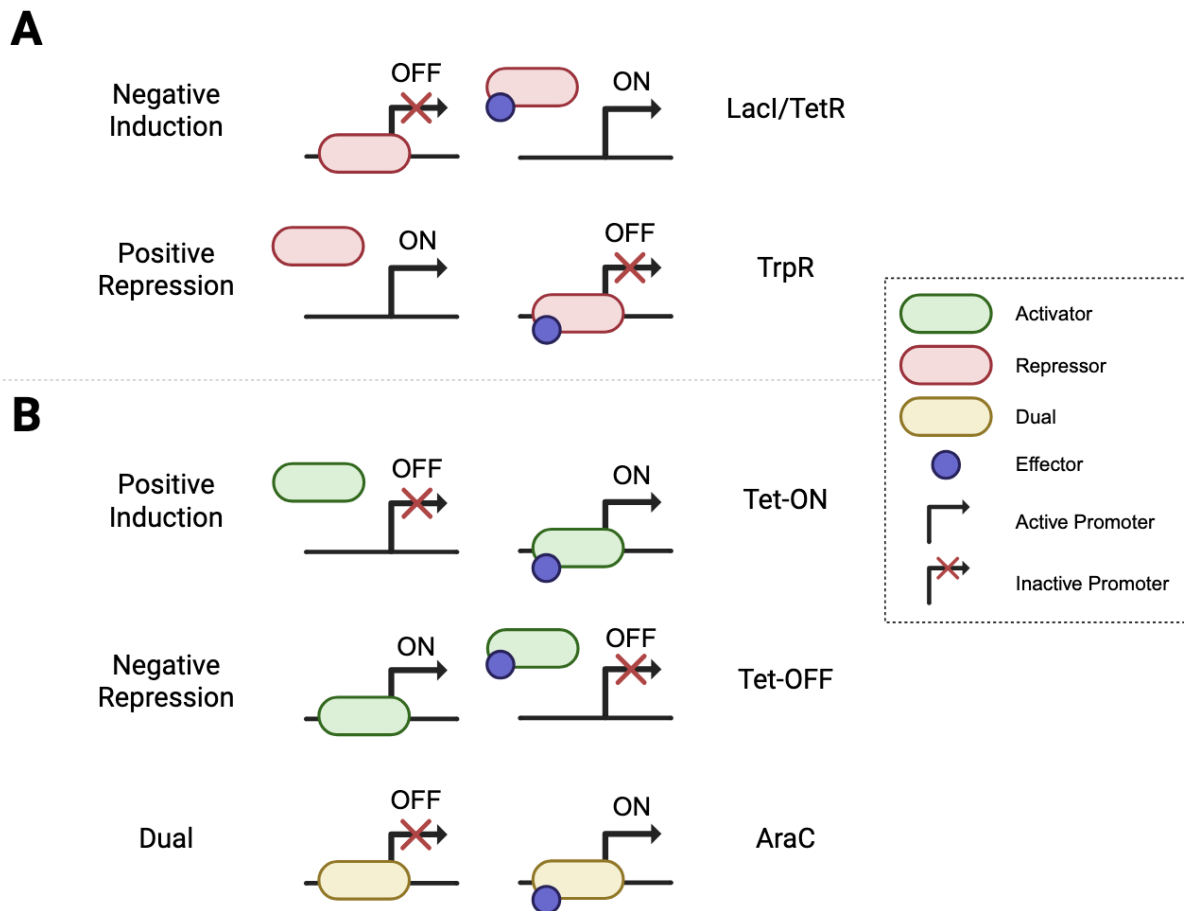


Figure 1. Regulatory roles of prokaryotic transcription factors. This scheme generally describes mechanisms of regulation seen at the level of transcription for prokaryotic operons. Green and red rounded rectangles represent transcription factors with activation (green) or repression (red) mechanisms. In the presence of their ligand, positive TFs bind to DNA, while negative TFs are released from it. Purple effectors represent activating molecules. At the right of each mechanism is a protein which exhibits this behaviour. **A:** Mechanisms exhibited by repressor TFs. Negative induction is also known as derepression. Binding of the TF to DNA represses transcription. **B:** Mechanisms exhibited by activator and dual TFs. Binding of the TF to DNA activates transcription. Dual TFs may either repress and activate while bound to DNA, depending on ligand availability. *Created in BioRender.*

1 – Introduction

A – General scientific context

Transcriptional regulation in synthetic biology

The regulation of gene expression is a fundamental aspect of cellular function, governing the precise control of protein production in response to environmental and intracellular signals. There are many different biological mechanisms for gene regulation at different stages of the central dogma. Notable examples are: genetic regulation through recombinase-mediated inversion, transcriptional regulation through DNA-binding transcription factor proteins (“TFs”), and translational regulation through RNA secondary structures (hairpins, riboswitches) or RNA interference. Transcriptional regulation is a prominent area of research, and many mechanisms relating the interaction between TFs, DNA binding sites, and RNA polymerase complexes have been revealed. Together, these components exhibit finely-tuned control over gene expression.

Prokaryotic transcription factors activate or repress gene expression

Prokaryotic transcription factors may exert control of gene expression through two possible mechanisms. These are (i) activation, where TF binding to an operator site helps form the RNA polymerase complex at the promoter, and (ii) repression, where TF binding physically prevents polymerase recruitment or blocks elongation (Fig. 1). Often, this activity is modulated by small molecular ligands that induce conformational changes in a TF, altering its affinity for its operator binding site. This behaviour gives rise to positive and negative systems, where transcription is switched either ON (1) or OFF (0) in response to a signal (Wang et al., 2011). For example, the LacI repressor loses affinity for its cognate operator upon binding allolactose, which derepresses the lac promoter (Lewis, 2005); the TetR/tTA/rtTA transcription factors are controlled by tetracycline, enabling the widely used Tet-ON and Tet-OFF systems (Zhou et al., 2006); LuxR acts as a transcriptional activator only upon binding acyl homoserine lactone (Yang et al., 2009); and TrpR represses transcription when bound to tryptophan (Merino et al., 2008). In addition, not all TFs are able to bind to a ligand. Some repressors act constitutively, so their regulatory effect is determined solely by their intracellular concentration and half-life.

Promoter abstraction enables construction of synthetic circuits

The predictable nature of TF-operator interactions has provided synthetic biologists with a growing repertoire of genetic parts with which they can engineer novel cellular behaviors.

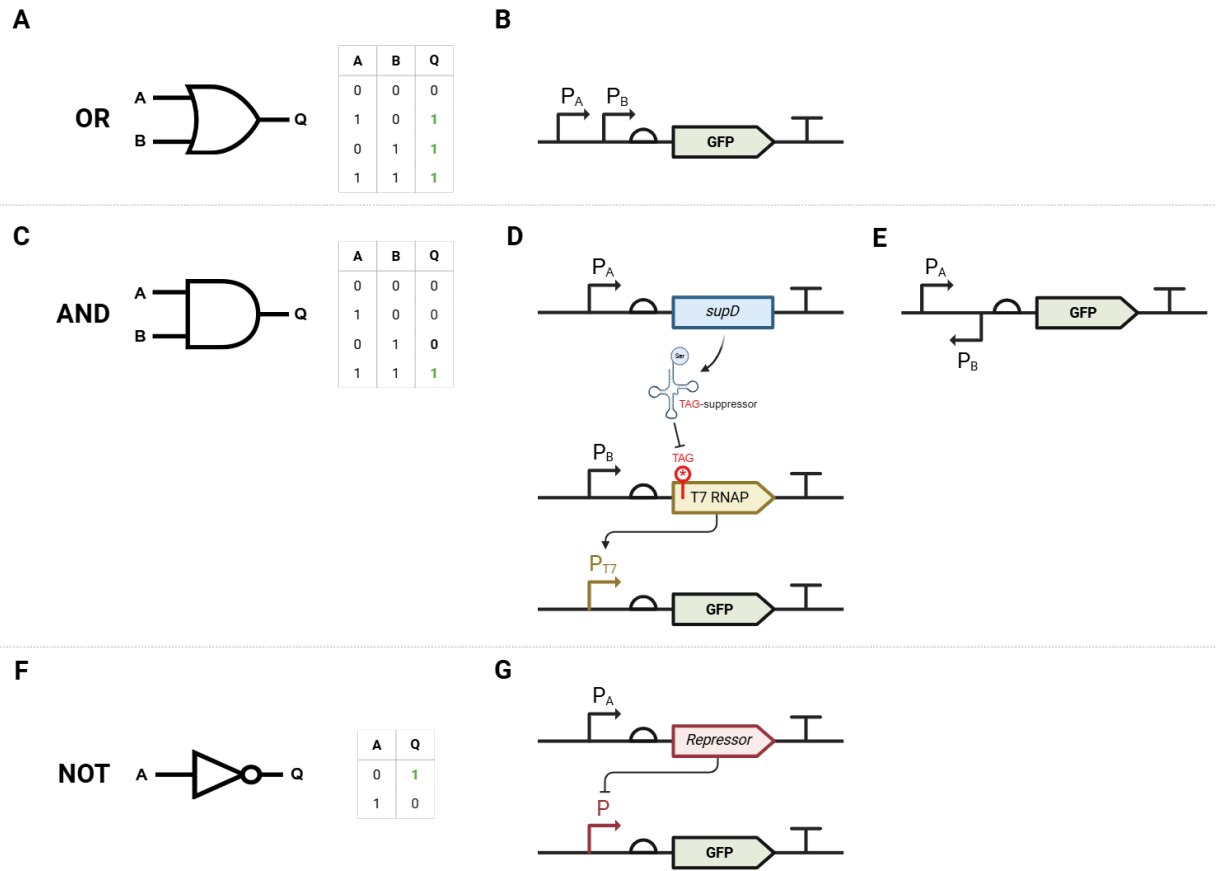


Figure 2. Primitive logic gates achievable using transcription factors. **A:** OR gate symbol and truth table. **B:** general promoter arrangement of an OR gate within a gene. **C:** AND gate symbol and truth table. **D:** Amber suppression AND gate which utilizes a anti-TAG tRNA^{Ser} under P_A and a T7 RNA Polymerase gene with early amber stop codon (TAG). **E:** general promoter arrangement of an AND gate with “subtractive” topology. P_A is a positive activator, and P_B is a positive repressor or negative inducer. **F:** NOT gate symbol and truth table. **G:** NOT gate implementation within a genetic circuit. Red “repressor” gene is a constitutively active repressor. *A and B: inputs; Q: output. Truth table: 0 = absent/inactive; 1 = present/active. P_A and P_B represent promoters responsive to inputs A and B. Synthetic Biology Open Language (SBOL) symbols: half-circle: ribosome binding site; bent arrow: promoter; T: terminator; pointed rectangle: protein coding sequence (Quinn et al., 2015). Created in BioRender.*

Early works have demonstrated that regulatory components could be abstracted from their native context and recombined to produce predictable, non-natural functions (Müller et al., 2025). Moreover, many efforts have been made to evaluate the functional characteristics of transcription factor binding sites, not only allowing creation of chimeric promoters such as P_{trc10} but also their *de novo* prediction through computational approaches. Conventionally, circuits in computer science are described by a topology of connected logic gates, inputs, and controls which give rise to complex functions. In synthetic biology, this concept was adapted to describe the use of different promoters to produce a desired outcome in response to various input stimuli; in essence, logical and reproducible expression of genes. At a higher level, tools such as CELLO enable the automated *in silico* design of complex logic circuits from characterized genetic parts (Nielsen et al., 2016). Synthetic toggle switches, repressilators, and logic gates all contribute to a rich and growing library of circuits for programming cellular decision-making.

Topologies and mechanisms of logic gates

The most straightforward logic gate is the OR gate, which is active when one or both of the two compared inputs are present (Fig. 2A). Normally, this is exhibited by the placement of two promoters that are each able to induce expression of a gene (Fig. 2B). The construction of OR gates using this topology is contingent upon the upstream promoter's ability to "pass through" the second downstream promoter. For example, as P_{BAD} is both a transcription activator and a repressor, occupation of the P_{BAD} operator in the absence of arabinose will block activity of any upstream promoter (Tamsir et al., 2011).

An AND gate produces an output only in the presence of both input signals (Fig. 2C). The commonly described topologies of this circuit include use of two transcriptional components that are codependent. Subtractive AND gates utilize two opposed promoters which compete for transcription activation. A viral T7 RNA polymerase gene with early amber stop codons that only produces functional polymerase in the presence of an amber-suppressor transfer RNA that is charged with an amino acid to continue translation can control the formation of the RNA polymerase complex on a gene with the T7 promoter (Fig. 2D). By having the sense promoter positively activated and the antisense promoter negatively repressed (such as *via* Tet-OFF), transcription only occurs when there is activation of P_A and repression of P_B (Fig. 2E). While the former topology is demonstrated and successful in the literature, it leaves very little room for more complex designs involving different gates interacting with each other.

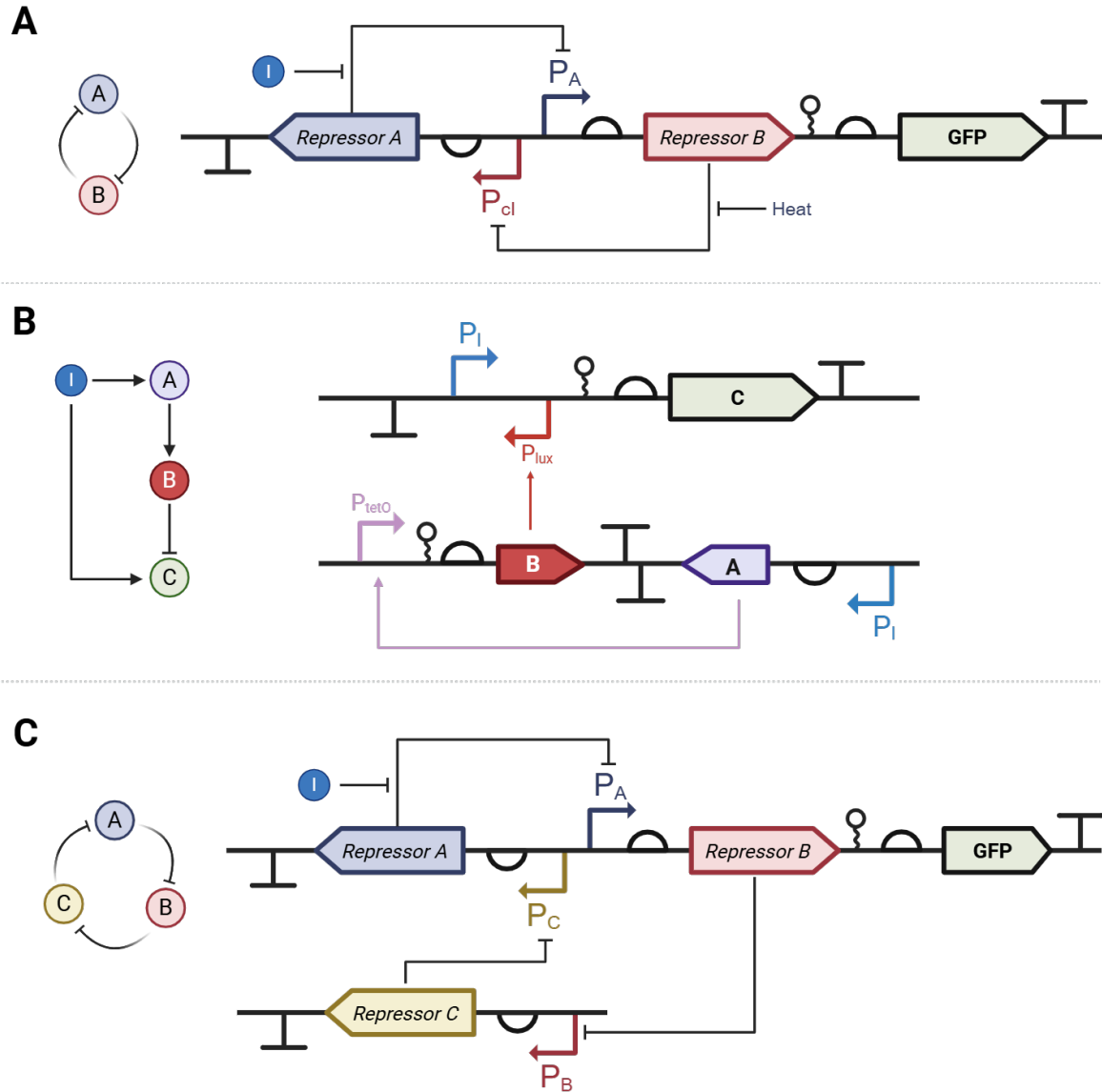


Figure 3. Topologies and genetic construction of representative dynamic genetic circuits.

A: Bistable switch composed of repressors A and B. Specifically, P_{cl} is repressed by Repressor B, and inducible by heat shock. P_A is repressed by Repressor A, and inducible by an effector "I" (blue circle). P_A transcribes an mRNA which contains a self-cleaving ribozyme insulator, thus co-expressing Repressor B and GFP reporter genes. **B:** Incoherent feedforward loop that is responsive to effector "I" (blue circle) via promoter P_A . Transcription factor A activates expression of factor B which is positioned antisense to P_A . This has a repression effect on the expression of C, the final product. **C:** Repressilator composed of three repressors which repress in series. When P_A is active, the reporter is also expressed, exhibiting an oscillatory behaviour. SBOL symbols as in Fig. 2; pin with wavy line: mRNA insulator (self-cleaving ribozyme). Created in BioRender.

NAND and NOR gates are functionally complete. Both AND and OR gates can be negated using a NOT gate (Fig. 2F), forming NAND and NOR gates. This is achieved by placing the next component under a normally-ON repressible promoter. NAND and NOR gates are unique in their ability to be “functionally complete”; in short, able to replicate the functionality of any other logic gate. Negation is a simple task in biological circuits, effected by the production of a constitutive repressor TF which inhibits expression of the output gene. Thus, achieving this in practice not only depends on the ability to negate any gate, but on ensuring that the components used can be freely combined without cross-talk.

Dynamic genetic circuits. Beyond replicating Boolean logic gates, synthetic gene circuits can be designed to exhibit dynamic temporal behaviors using feedback and feedforward loops. Bistable switches exploit mutual repression between two transcriptional units to produce a system with two stable steady states (Fig. 3A) (Huan and Wang, 2024). Oscillators such as the repressilator extend this principle to three or more (in odd numbers) mutually repressing components, exhibiting periodic gene expression whose periods are determined by the degradation rates and expression levels of their constituent TFs (Elowitz and Leibler, 2000). Incoherent feedforward loops, where an input signal simultaneously activates an output and induces a delayed repression of that same output, produce an ephemeral pulse response to a sustained input; useful for input rising edge detection and filtering out transient signals (Fig. 3B) (Litovco et al., 2021). Repressilators are oscillating circuits which contain an odd number of mutually repressed genes (Fig. 3C). These are able to create periodic intervals of gene expression, applicable as an internal clock.

Cloning methods and assembly standards

The construction of synthetic gene circuits requires the precise assembly of multiple genetic parts in a specific order, a task that becomes increasingly complex as circuit size and complexity grows. Early cloning methods relying on conventional Type II restriction enzymes and ligation were limited by the scarcity of unique restriction sites and the incompatibility of fragments generated by different enzymes, making the assembly of multi-part constructs laborious and stochastic. Moreover, as these enzymes recognize palindromic sequences, cut sites or scars typically remain in the final construct. The usage of Type IIS (shifted) restriction enzymes that cleave DNA asymmetrically at a fixed distance from their recognition site proposes a solution to this problem. They enable the

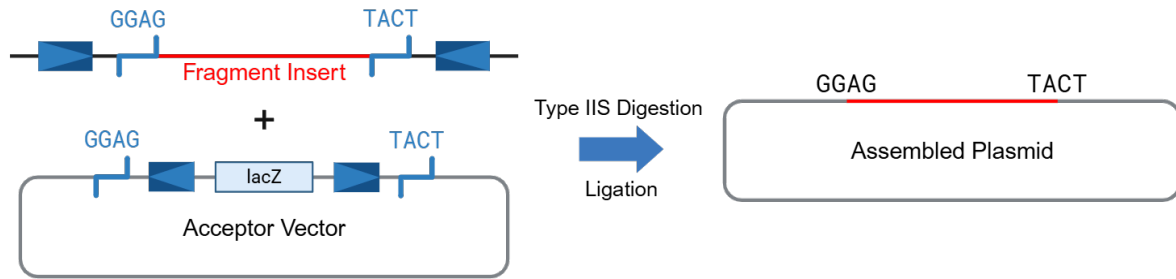


Figure 4. Golden Gate Assembly of a single insert into an acceptor vector. Blue boxes correspond to directional cut sites of a Type IIS restriction enzyme such as BsaI, BsmBI, and BbsI. The 5' and 3' overhangs match, and are able to assemble together in a restriction-ligation reaction. Restriction binding sites are removed, and a scarless ligation is achieved. This method can be extended to many fragments in one construction. *Created in BioRender.*

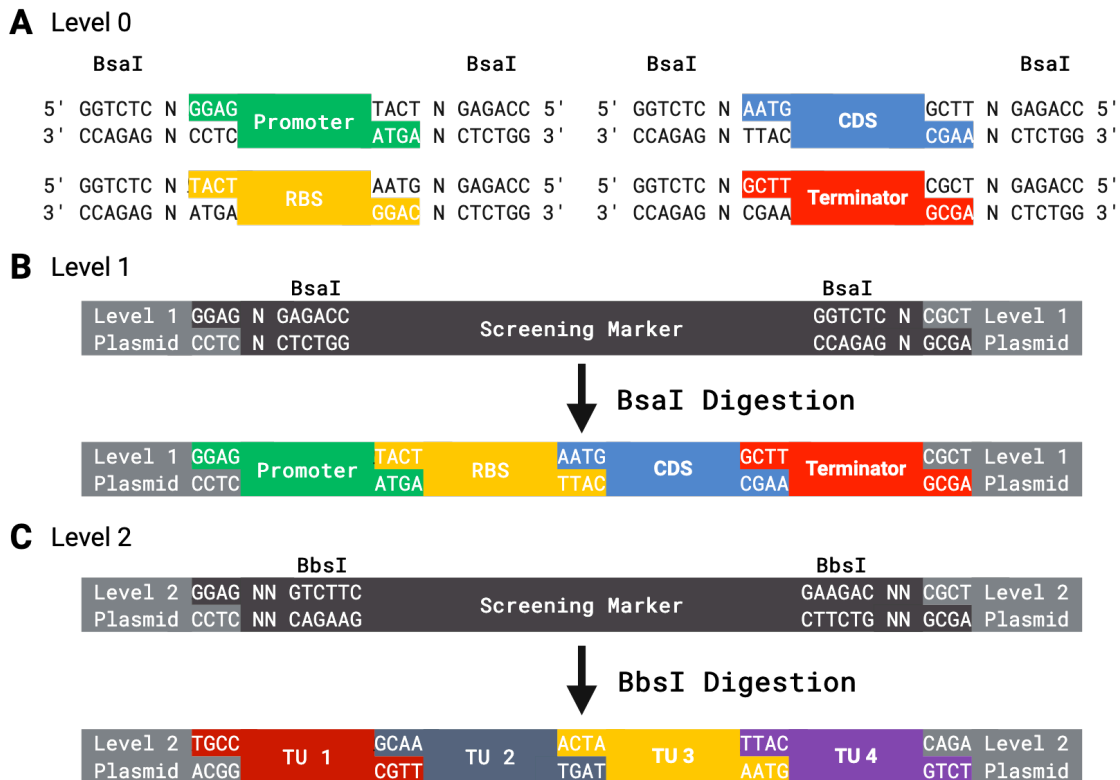


Figure 5. Illustration of the MoClo hierarchical cloning scheme. This scheme shows the MoClo cloning workflow which uses the BsaI and BbsI Type IIS restriction enzymes. **A:** generalized overhang choice for individual Level 0 parts. **B:** Level 1 transcriptional unit (TU) assembly using Level 0 parts and BsaI. At Level 0 construction, Loop and MoClo share overhang design and assembly restriction enzyme. **C:** MoClo Level 2 assembly using Level 1 TUs and BbsI. *Adapted from https://parts.igem.org/Help:Standards/Assembly/Type_IIS. Created in BioRender.*

generation of rational, defined DNA overhangs outside the recognition sequence. This property is exploited by Golden Gate Assembly (hereafter GGA), during which multiple fragments ligate in a reproducible order with no scar sequences at the junctions (Fig. 4). This method also enables one-pot reactions using only one restriction enzyme with ligase and DNA, reducing the likelihood that sequences require domestication (removal of illegal internal cut sites).

Hierarchical DNA assembly

Building upon GGA, the Modular Cloning (MoClo) standard comprises a framework for hierarchical assembly. First, linear DNA sequences are adapted using PCR or DNA synthesis and subcloned into “Level –1” vectors. These basic genetic sequences are then inserted into a Level 0 plasmid by GGA and formalized as “Level 0” parts; namely, promoters, ribosome binding sites (RBSs), coding sequences (CDSs), and terminators (Fig. 5A). Within the framework, Level 0 parts are given specific DNA overhangs. These are then combined into “Level 1” transcriptional units (Fig. 5B), which may stand alone or finally be assembled into multi-gene constructs as “Level 2” (Fig. 5C) (Weber et al., 2011). The Loop cloning standard enables further iterative cycles of assembly allowing for constructs unlimited in size (Pollak et al., 2019).

Generally, assembly frameworks contain formalized overhang sequences which flank different functional regions of a gene. These standards have been widely adopted in the synthetic biology community, enabling the reuse and exchange of parts between laboratories and organizations. Moreover, it allows for abstracting functions from their primary sequences, meaning that laboratories can use the same parts to assemble the same device and expect consistent results (Weber et al., 2011). The use of GGA-based assembly standards are supported by the iGEM¹ Foundation’s RFC[1000], which provides guidelines for interoperability.

Control over the promoter sequence within current frameworks is limited

While MoClo and related hierarchical systems have accelerated the construction of multi-gene devices, granularity at the promoter level is rather limited. Their treatment of the promoter as an indivisible unit limits the ability to fine-tune transcriptional inputs or rapidly swap regulatory elements across circuit architectures. Increasingly complex synthetic gene circuits demand an approach with finer modularity. In practice, the promoter region is a composite of interacting elements whose combinatorial arrangement determines the logical properties of

¹iGEM: International Genetically Engineered Machine

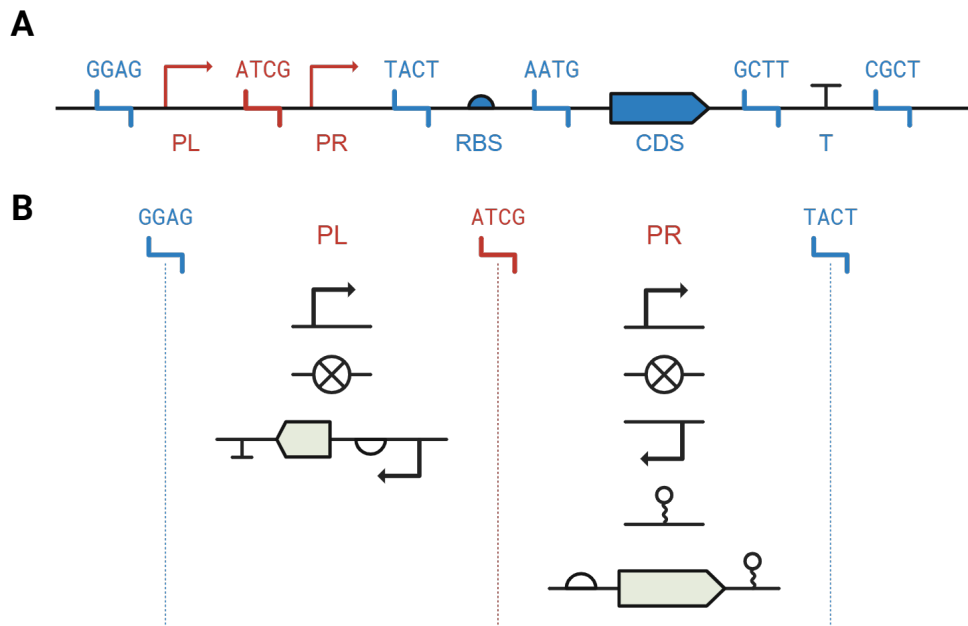


Figure 6. Scheme depicting the expanded MoClo standard with PL/PR architecture. **A:** MoClo standard overhangs (blue) with PL and PR parts and new ATCG overhang (red). Depicted parts, when assembled, would constitute a complete TU. **B:** Variants of Level 0 parts which may be inserted at the PL or PR regions of a constructed TU. This is not exhaustive, but representative of possible constructs for this project. *SBOL symbols as in (Figs. 2,3) circle with cross: spacer or dummy (Quinn et al., 2015). Created in BioRender.*

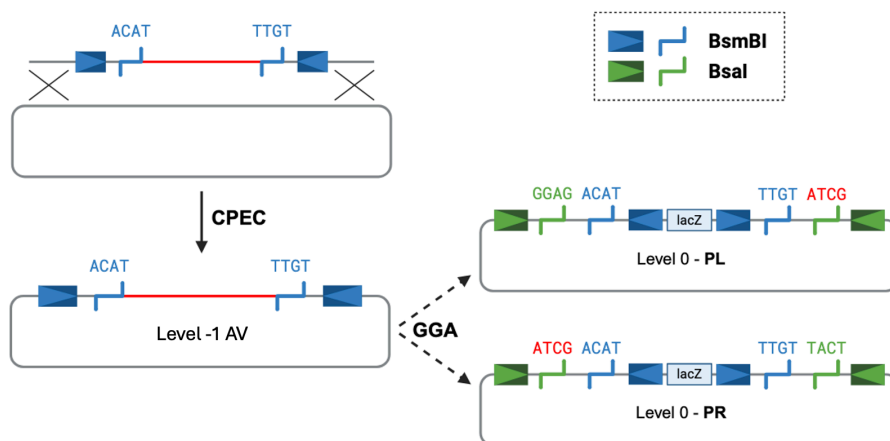


Figure 7. Workflow from DNA fragments to Level 0 parts through Level -1 fragments and plasmids. After PCR or synthesis, a fragment is cloned into pAGM1311 using CPEC. Then, using Golden Gate Assembly, the part can enter a Level 0 vector corresponding to either PL or PR. Crossed lines represent homologous regions of DNA.

gene expression. A cloning standard that treats these elements as independently exchangeable parts would enable circuit flexibility that is currently unavailable within the existing frameworks.

Design of the genetic circuit “promoter toolkit”

To address these limitations, the host lab conceptualized a genetic circuit “promoter toolkit.” The central design philosophy of this toolkit proposes the subdivision of the promoter region into two independently exchangeable regions: a left promoter (PL) and a right promoter (PR), separated by a novel DNA overhang at their junction (Fig. 6A). This modularity enables three powerful applications. The first is the embedding of a TF expression cassette within PL, such that a device becomes self-sufficient with respect to the regulatory interaction it participates in (Fig. 6B). This is particularly useful for heterologous promoters whose TF is not natively present in the host organism. The second application is the placement of two independent operator sequences within the promoter region, enabling dual-input transcriptional control within a single Level 0 part (Fig. 6B). This provides an architectural basis for the logic gates previously described of various different topologies. The third is the embedding of a TF expression cassette within PR that is under control of PL, and insulated from the downstream RBS and coding sequence using a self-cleaving ribozyme called RiboJ (Clifton et al., 2018) which removes the cis-regulatory region from the final mRNA (Fig. 6B). This enables the construction of bistable switches and oscillators which function via mutual regulatory dependencies between devices. Together, these three applications make the PL/PR architecture a flexible and compact platform for encoding complex regulatory logic directly at the promoter level.

In practice, this subdivision will be implemented by splitting the MoClo promoter functional region (demarcated by the 5' GGAG and 3' TACT fusion sites) into two parts using a defined internal junction site (Fig. 6A). Based on an analysis of four-nucleotide overhang ligation fidelities (Potapov et al., 2018), the sequence ATCG was chosen for the junction overhang due to its low mismatch annealing frequency to other existing MoClo overhangs and its comparable ligation efficiency, minimizing the risk of misassembly while keeping efficiency high. The resulting PL and PR parts are cloned as Level 0 parts flanked by their respective fusion sites (5'-GGAG|PL|ATCG-3' and 5'-ATCG|PR|TACT-3') and are therefore fully compatible with the existing MoClo hierarchy without modification to downstream assembly levels. The workflow for cloning a promoter fragment through Level -1 into either a PL or PR Level 0 acceptor vector is illustrated in Fig. 7.

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B – Scientific questions and objective of the project

The scientific question posed in this project is: “*Can a plasmid assembly standard with expanded functionality at the promoter region of a transcriptional unit enable the reliable construction and accurate dose-dependent response of synthetic gene circuits across diverse topologies and dynamic mechanisms?*”

Objective #1: Design a modular genetic circuit toolkit based on an expanded MoClo promoter assembly standard *in silico*

Both common and niche synthetic regulatory circuit topologies were selected from literature or designed *in silico* using inducible promoters; including P_{lac} , P_{tet} , P_{lux} , heavy metal-responsive, heat shock, and CELLO-described promoters (Jones et al., 2022; Nielsen et al., 2016). These are chosen for their ability to constitute logic gates, oscillators, and bistable switches. Can the breadth of these topologies be accommodated within a single unified modular framework, and does the PL/PR promoter split provide sufficient flexibility for combinatorial circuit design across all topologies?

Objective #2: Establish the reliability of the expanded MoClo standard through construction of Level –1, 0, and 1 genetic parts

The physical generation of the genetic toolkit will follow, utilizing *de novo* gene synthesis or PCR, and plasmid cloning to generate a library of Level –1 and Level 0 genetic parts, kept within various housekeeping plasmids. Moreover, the design of the PL/PR split promoter system will be subject to assembly of Level 1 transcriptional units and analyzed by PCR and next-generation sequencing. Does the ATCG junction perform with sufficient fidelity for reliable assembly, and can the expanded standard produce these constructs consistently?

Objective #3: Characterize the logical and dynamic behavior of assembled synthetic gene circuits *in vivo*

Level 1 and 2 test devices assembled from the toolkit will be characterized in *E. coli* NEB 10-beta using fluorescent protein reporters, measuring steady-state expression levels, dose-response curves, and kinetics across a range of effector concentrations. Do assembled circuits produce theoretically predicted outputs, and do circuits implementing the same logical function through different molecular topologies produce comparable dynamic behaviors?

Table 1. Primers designed and used within this work. † PS1-PS6 primers are standardized as a part of the pSEVA architecture (Martínez-García et al., 2023).

Name	Sequence
PS1_T1fwd †	AGGGCGGCGGATTTGTCC
PS2_T0rev †	GCGGCAACCGAGCGTTC
PS3_T0fwd †	GAACGCTCGGTTGCCGC
PS4_OriTrev †	CCAGCCTCGCAGAGCAGG
PS5_OriTfwd †	CCCTGCTTCGGGGTCATT
PS6_T1rev †	GGACAAATCCGCCGCCCT
iG24-030_CmRmut-R	ataCGTCTCaCACGaaaacatattctcaataaaccttaggg
iG24-031_CmRmut-F	ataCGTCTCaCGTgtcagccaatccctggg
iG26-036_PS8-CloDF13ori	cctgctctgcgaggctggcgtaggccggccTCACTCGGTCGCTACGCTCC
iG26-037_PS6-CloDF13ori	ggacaaatccgccccctagacagctggcgccgcccGCGGTTCACTAGAAAAGATCAAAGGATCTTCTTGAGATCC
iG26-038_pSB1C5SA-Fwd	TACTTGAGACCTATAAACGCCAGGTTGTATCC
iG26-039_pSB1C5SA-Rev	CTCCCGAGACCGAACCCGAGC
iG26-040_pSB1C5SAL-Fwd	ATCGTGAGACCTATAAACGCCAGGTTGTATCC
iG26-041_pSB1C5SAR-Rev	CGATCGAGACCGAACCCGAGC
iG26-042_pSB1C5SB-Fwd	AATGTGAGACCTATAAACGCCAGGTTGTATCC
iG26-043_pSB1C5SB-Rev	AGTACGAGACCGAACCCGAGC
iG26-044_pSB1C5SC-Fwd	GCTTTGAGACCTATAAACGCCAGGTTGTATCC
iG26-045_pSB1C5SC-Rev	CATTCGAGACCGAACCCGAGC
iG26-046_pSB1C5SD-Fwd	CGCTTGAGACCTATAAACGCCAGGTTGTATCC
iG26-047_pSB1C5SD-Rev	AAGCCGAGACCGAACCCGAGC
iG26-048_J23104-L0-Fwd	cattgctcgggttcggtctcgggagTTGACAGCTAGCTCAGTCCTAGG
iG26-049_B0011-L0-Rev	gcgtttataggtctcaagcAAATAATAAAAAAGCCGGATTAATAATCTGG
iG26-050_XylS-L0-Rev	gcgtttataggtctcaagcTCAAGCCACTTCCTTTTTGCATTGACGC
iG26-051_PsiR-L0-Fwd	cattgctcgggttcggtctcgaATGACCGGTATCTCTTTCTAAAAAAGCTACC
iG26-052_PsiR-L0-Rev	gcgtttataggtctcaagcTTATTTAGCCATACGACGCAGAGC
iG26-053_P_BAD-L0-Fwd	cattgctcgggttcggtctcgggagACTTTTCATACTCCCGCCATTGAGAGAAGAAACC
iG26-054_PR_BAD-L0-Fwd	cattgctcgggttcggtctcgaATGACCGGTATCTCTTTCTAAAAAAGCTACC
iG26-055_P_BAD-L0-Rev	gcgtttataggtctcaagcGCTAGCCCCAAAAAACGGGTATGG
iG26-056_PL_BAD-L0-Rev	gcgtttataggtctcagcGCTAGCCCCAAAAAACGGGTATGG
iG26-057_LacI-AM-L0-Fwd	cattgctcgggttcggtctcgaATGAAACAGTAACGTTATACGATGTCG
iG26-058_LacI-AM-L0-Rev	gcgtttataggtctcaagcTCACTGCCCGCTTTCCAGTCG
iG26-059_AraC-L0-Fwd	cattgctcgggttcggtctcgaATGGCTGAAGCGCAAATGATCC
iG26-060_AraC-L0-Rev	gcgtttataggtctcaagcTTATGACAACTTGACGGCTACATCATTAC
iG26-061_pSB1C5Sx-Rev	CATTGCTCGGGTTCGGTCTCG
iG26-062_pSB1C5Sx-Fwd	GCGGATACAACCTGGCGTTTATAGG
iG26-067_sfGFP-L0-Fwd	cattgctcgggttcggtctcgaATGCGTAAAGCGAAGAACTGTTTACC
iG26-068_LVAtag-L0-Rev	gcgtttataggtctcaagcTTATTAAGCTACTAAAGCGTAGTTTTCGTCG

2 – Materials and Methods

In silico design

All part sequences were designed and catalogued using SnapGene (Dotmatics, Boston, MA). Golden Gate overhang compatibility and potential misassembly were verified *in silico* prior to ordering or amplification. Furthermore, all primers were designed manually in SnapGene. All sequences were screened for internal illegal BsaI and BsmBI/Esp3I recognition sites. To remove illegal sites, primers were found to remove them using GGA, where a mutation is introduced in the variable overhang generated by Type IIS restriction enzymes.

Generation of linear genetic parts

All PCR amplifications were performed using Q5 High-Fidelity DNA Polymerase (New England Biolabs, MA, USA) with primers designed to append the appropriate 4-nucleotide GGA overhangs and CPEC homology at each terminus. The iGEM 2024 and 2025 distribution kits provided the following templates: BBa_I0500 for P_{BAD} and AraC, and BBa_K2680537 for LacI-AM. An in-house plasmid was used as template for sfGFP-LVA (BBa_K2675006), PsiR CDS (BBa_K2448006), and XylS (BBa_K5061060) transcriptional unit were amplified from in-house plasmid templates. Primers are listed in Table 1. PCR reactions were carried out in a final volume of 25 μ L containing 1X Q5 Master Mix, 0.5 μ M of each primer and 0.1 ng of DNA template. Thermocycling consisted of initial denaturation at 98°C for 30 sec, followed by 35 cycles of 98°C for 10 sec, 30 sec at annealing temperature, and 72°C extension (30 seconds per kb). Annealing temperature was chosen with the help of the NEB Tm Calculator. All PCR products were verified by 1% agarose gel electrophoresis using 4 μ L of product. (Midori Green; NIPPON Genetics EUROPE. 1 kb Plus Ladder; New England Biolabs).

Promoters P_{tacI} , $P_{PsiTac1}$, and P_m (and PL/PR versions), as well as synthetic RBS and terminator sequences, were ordered as double-stranded synthetic gene fragments (gBlocks, IDT, Coralville, IA) and used directly in assembly reactions (Table 2).

Backbone vectors

To eliminate an internal BsmBI recognition site within the CmR gene of pSB1C5, whole-plasmid PCR was performed using primers iG24-030 and iG24-031 (Table 1). These primers amplify the entire plasmid while omitting the BsmBI site, and encode the C435G point mutation within the

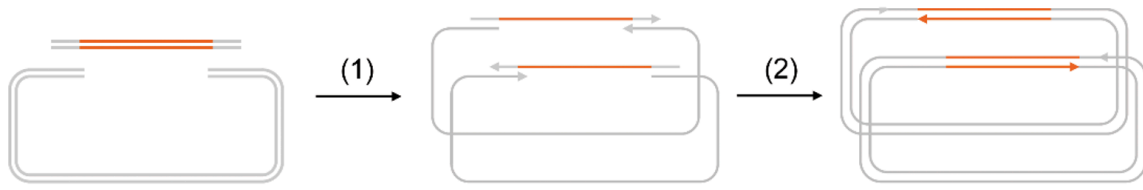


Figure 8. Scheme describing circular polymerase extension cloning (CPEC). DNA is first denatured, then cooled to allow for DNA annealing (1). Then, an extension step (2) performed by DNA polymerase completes the other strand. Adapted from Quan and Tian (2009).

Table 2. Commercially synthesized linear DNA fragments used in this project. All sequences are capable of CPEC assembly into the pSB1C5(BsmBI-free) AV, and in GGA reactions.

Fragment Name	Description	Reference
iG26-GB001_P-RBS-J23101-RBS25kPsiR	BBa_J23101 + synthetic PsiR RBS	Anderson (2006); Salis et al. (2009)
iG26-GB002_P-RBS-J23104-RBS25kAraC	BBa_J23104 + synthetic AraC RBS	Anderson (2006); Salis et al. (2009)
iG26-GB003_P-RBS-LacIq	LacIQ Promoter + RBS	Calos (1978)
iG26-GB004_pL0c-T_L2U3H03	Synthetic Terminator	Chen et al. (2013)
iG26-GB005_pL0c-T_L3S1P51	Synthetic Terminator	Chen et al. (2013)
iG26-GB006_pL0c-T_L3S3P22	Synthetic Terminator	Chen et al. (2013)
iG26-GB007_pL0c-P_Pm	P _m Promoter	BBa_K3202001
iG26-GB008_pL0c-P_pPsiTacl	P _{PsiTacl} Promoter	BBa_K2448016
iG26-GB009_pL0c-P_Tacl	P _{tacl} Promoter	Boer et al. (1983)
iG26-GB010_pL0c-PL_Pm	PL _m Promoter	BBa_K3202001
iG26-GB011_pL0c-PL_pPsiTacl	PL _{PsiTacl} Promoter	BBa_K2448016
iG26-GB012_pL0c-PL_Tacl	PL _{tacl} Promoter	Boer et al. (1983)
iG26-GB013_pL0c-PR_Pm	PR _m Promoter	BBa_K3202001
iG26-GB014_pL0c-PR_pPsiTacl	PR _{PsiTacl} Promoter	BBa_K2448016
iG26-GB015_pL0c-PR_Tacl	PR _{tacl} Promoter	Boer et al. (1983)
iG26-GB016_RiboJ-K2675017-T30A	RiboJ insulator + sfGFP RBS	Lou et al. (2012); BBa_K2675017
iG26-GB017_pL0c-T_L3S2P56	Synthetic Terminator	Chen et al. (2013)
iG26-GB018_pL0c-T_L3S2P21	Synthetic Terminator	Chen et al. (2013)
iG26-GB019_JUMP-1A-LacZ	pJUMPxx-1A GGA payload	Valenzuela-Ortega and French (2021)
iG26-GB020_JUMP-1AB-LacZ	pJUMPxx-1AB GGA payload	This study
iG26-GB021_JUMP-1B-LacZ	pJUMPxx-1B GGA payload	Valenzuela-Ortega and French (2021)

BsmBI-generated overhang, such that BsmBI digestion produces a CGTG overhang carrying the mutation. The linear PCR product was purified and re-circularized by self-ligation in a BsmBI GGA reaction under the conditions described below. Transformants were selected on LB-agar supplemented with chloramphenicol.

Plasmid construction

CPEC assembly

CPEC is a method of DNA assembly that does not use restriction enzymes. Instead, it uses DNA polymerase to join two or more DNA molecules by subsequent melting, annealing, and extension steps in a thermal cycler (Fig. 8) (Quan and Tian, 2009; Quan and Tian, 2011).

An initial attempt to construct the pJUMP4x and pJUMP7x backbone plasmids utilized CPEC (Quan and Tian, 2009; Quan and Tian, 2011). Three linear fragments were amplified: the antibiotic resistance cassette from the respective pSEVA vector, the CloDF13 ORI from MP6, and the JUMP payload architecture from various pJUMP29-1x plasmids recovered from the 2024 iGEM Distribution Kit. Primers and templates used for each reaction can be found in Table 3. CPEC reactions were performed using Q5 polymerase, with an initial denaturation at 98 °C for 30 seconds, followed by 9 cycles of denaturation (98 °C, 10 seconds), annealing (65 °C, 15 seconds), and extension (72 °C, 2 minutes), a final extension at 72 °C for 5 minutes, and a hold at 16 °C. Upon encountering unexpected results, the JUMP payload fragment was replaced with a commercially synthesized DNA fragment encoding the required architecture (Table 2), and the three-fragment CPEC assembly was repeated.

Golden Gate Assembly

Level 1 test device assembly. Each reaction contained the pJUMP27-1A acceptor vector, one or two promoter fragments (P or PL and PR), a synthetic RBS fragment, the sfGFP-LVA CDS, and a synthetic terminator fragment, assembled using BsaI-HFv2 (New England Biolabs). **Level 1 accessory expression plasmid assembly.** Each TF expression cassette (Anderson promoter::RBS::TF::terminator) was assembled into the appropriate pJUMP4x (1A or 1B) backbone by BsaI GGA under identical conditions. **Level 2 assembly.** Dual-TF Level 2 constructs were assembled by Esp3I (BsmBI isoschizomer; Thermo Fisher Scientific, MA, USA) GGA using the Level 1 TU inserts and the pJUMP7x-1AB backbone under the same program.

Table 3. CPEC assemblies to be performed with primers for each reaction. *SmR*: spectinomycin resistance; *TmpR*: trimethoprim resistance; AV: acceptor vector.

Plasmid Name	AbR	CPEC Fragments	Type
pJUMP4x-1D'	SmR	1. PCR PS1 & PS2 on pJUMP29-1D'(sfGFP) 2. PCR PS3 & PS4 on pSEVA481 3. PCR iG26-036 & iG26-037 on MP6	Level 1 AV
pJUMP4x-1E	SmR	1. PCR PS1 & PS2 on pJUMP29-1E(sfGFP) 2. PCR PS3 & PS4 on pSEVA481 3. PCR iG26-036 & iG26-037 on MP6	Level 1 AV
pJUMP7x-1D	TmpR	1. PCR PS1 & PS2 on pJUMP29-1D(sfGFP) 2. PCR PS3 & PS4 on pSEVA721 3. PCR iG26-036 & iG26-037 on MP6	Level 2 AV
pJUMP4x-1A(LacZ)	SmR	1. iG26-GB019_JUMP-1A-LacZ 2. PCR PS3 & PS4 on pSEVA481 3. PCR iG26-036 & iG26-037 on MP6	Level 1 AV
pJUMP4x-1B(LacZ)	SmR	1. iG26-GB021_JUMP-1B-LacZ 2. PCR PS3 & PS4 on pSEVA481 3. PCR iG26-036 & iG26-037 on MP6	Level 1 AV
pJUMP7x-1AB(LacZ)	TmpR	1. iG26-GB020_JUMP-1AB-LacZ 2. PCR PS3 & PS4 on pSEVA721 3. PCR iG26-036 & iG26-037 on MP6	Level 2 AV

All Golden Gate reactions were run in a total volume of 20 μ L, beginning with a 20-minute incubation at 37 °C, followed by 15 cycles of digestion and ligation (1 minute 30 seconds at 37 °C, then 3 minutes at 16 °C), a final 5-minute step at 50 °C, and enzyme inactivation at 80 °C for 10 minutes, before holding at 16 °C. After assembly, 3 μ L of each reaction was transformed into 25 μ L chemically competent NEB 10-beta cells. Transformants were plated on LB-agar supplemented with the appropriate antibiotic and X-Gal (Sigma-Aldrich, MO, USA) then incubated overnight at 37 °C.

Colony PCR verification

Following transformation, 4–8 colonies per assembly reaction were screened by colony PCR using primers flanking the predicted assembly junctions. For this, bacteria from an isolated colony were first boiled in 20 μ L of water for 3 minutes at 95°C, then 1 μ L was used as template in a PCR reaction (20 μ L) containing 10 μ L of DreamTaq Master Mix 2X (Thermo Fisher Scientific) and 1 μ L of each of the two primers at 10 μ M. Thermocycling consisted of an initial denaturation at 95°C for 3 min, followed by 35 cycles of 95°C for 30 s, 63°C for 30 s, and 72°C for 1 min/kb. Amplicons (7 μ L) were analyzed on a 1% agarose gel, and clones yielding a band of the expected size were selected for liquid culture and subsequent plasmid extraction.

Plasmid DNA isolation, analytical digestion and Nanopore sequencing

Plasmid DNA was extracted from overnight cultures of colony PCR-positive clones using Monarch DNA Purification Kit (New England Biolabs) for 5 mL cultures or PureYield plasmid Midiprep kit (Promega, WI, USA) for 50-100 mL cultures..

Analytical digestions were performed on 100 ng of plasmid DNA with 1 μ L of each FastDigest restriction enzyme (Thermo Fisher Scientific) in FastDigest Green Buffer (15 μ L total volume). After 1 hour at 37°C, reactions were loaded directly onto a 1.0% agarose gel

Whole-plasmid sequencing was performed by Oxford Nanopore sequencing by Eurofins (Oxford Nanopore Technologies, Oxford, UK).

Table 4. Golden Gate Assemblies to be performed. Enz.: restriction enzyme; KanR: kanamycin resistance; SmR and TmpR as in Table 3. Plasmid naming convention: p[Level]<Position>[AbR(c/s/t)]-<P/PL/PR/T>_PartName

Plasmid	AbR	Enz.	Backbone	Fragment 1	Fragment 2	Fragment 3	Fragment 4	Fragment 5
pL1k_pTacl_RibolJ-K2675017-T30_sfGFP_L3S2P56	KanR	Bsal	pJUMP27-1A	iG26-GB009_pL0c-P_Tacl	iG26-GB016_RibolJ-K2675017-T30A	PCR iG26-067 & iG26-068 on BBa_K5061056	iG26-GB017_pL0c-T_L3S2P56	—
pL1k_pBad_RibolJ-K2675017-T30_sfGFP_L3S2P56	KanR	Bsal	pJUMP27-1A	PCR iG26-053 & iG26-055 on BBa_I0500	iG26-GB016_RibolJ-K2675017-T30A	PCR iG26-067 & iG26-068 on BBa_K5061056	iG26-GB017_pL0c-T_L3S2P56	—
pL1k_pBad_pLac_RibolJ-K2675017-T30_sfGFP_L3S2P56	KanR	Bsal	pJUMP27-1A	PCR iG26-053 & iG26-056 on BBa_I0500	iG26-GB015_pL0c-PR_Tacl	iG26-GB016_RibolJ-K2675017-T30A	PCR iG26-067 & iG26-068 on BBa_K5061056	iG26-GB017_pL0c-T_L3S2P56
pL1k_pLac_pBad_RibolJ-K2675017-T30_sfGFP_L3S2P56	KanR	Bsal	pJUMP27-1A	iG26-GB012_pL0c-PL_Tacl	PCR iG26-054 & iG26-055 on BBa_I0500	iG26-GB016_RibolJ-K2675017-T30A	PCR iG26-067 & iG26-068 on BBa_K5061056	iG26-GB017_pL0c-T_L3S2P56
pL1k_Pm_RibolJ-K2675017-T30_sfGFP_L3S2P56	KanR	Bsal	pJUMP27-1A	iG26-GB007_pL0c-P_Pm	iG26-GB016_RibolJ-K2675017-T30A	PCR iG26-067 & iG26-068 on BBa_K5061056	iG26-GB017_pL0c-T_L3S2P56	—
pL1k_pPsiTacl_RibolJ-K2675017-T30_sfGFP_L3S2P56	KanR	Bsal	pJUMP27-1A	iG26-GB008_pL0c-P_pPsiTacl	iG26-GB016_RibolJ-K2675017-T30A	PCR iG26-067 & iG26-068 on BBa_K5061056	iG26-GB017_pL0c-T_L3S2P56	—
pL1k_Pm_pPsiTacl_RibolJ-K2675017-T30_sfGFP_L3S2P56	KanR	Bsal	pJUMP27-1A	iG26-GB010_pL0c-PL_Pm	iG26-GB014_pL0c-PR_pPsiTacl	iG26-GB016_RibolJ-K2675017-T30A	PCR iG26-067 & iG26-068 on BBa_K5061056	iG26-GB017_pL0c-T_L3S2P56
pL1k_pPsiTacl_Pm_RibolJ-K2675017-T30_sfGFP_L3S2P56	KanR	Bsal	pJUMP27-1A	iG26-GB011_pL0c-PL_pPsiTacl	iG26-GB013_pL0c-PR_Pm	iG26-GB016_RibolJ-K2675017-T30A	PCR iG26-067 & iG26-068 on BBa_K5061056	iG26-GB017_pL0c-T_L3S2P56
pL1As_AraC_L3S2P21	SmR	Bsal	pJUMP4x-1A	iG26-GB002_P-RBS-J23104-RBS25kAraC	PCR iG26-059 & iG26-060 on AraC plasmid (p109?)	iG26-GB018_pL0c-T_L3S2P21	—	—
pL1As_XylS_B0011	SmR	Bsal	pJUMP4x-1A	XylS(TU) PCR iG26-048 & iG26-049 on p328	—	—	—	—
pL1Bs_LacI-AM_L3S1P51	SmR	Bsal	pJUMP4x-1B	iG26-GB003_P-RBS-LacIq	PCR iG26-057 & iG26-058 on pL0c-LacI-AM	iG26-GB005_pL0c-T_L3S1P51	—	—
pL1Bs_PsiR_L3S1P51	SmR	Bsal	pJUMP4x-1B	iG26-GB001_P-RBS-J23101-RBS25kPsiR	PCR iG26-057 & iG26-058 on G66 (iGEM 2018?)	iG26-GB005_pL0c-T_L3S1P51	—	—
pL2L_AraC_LacI-AM	TmpR	BsmBI	pJUMP7x-1AB	pL1As_AraC_L3S2P21	pL1Bs_LacI-AM_L3S1P51	—	—	—
pL2L_XylS_PsiR	TmpR	BsmBI	pJUMP7x-1AB	pL1As_XylS_B0011	pL1Bs_PsiR_L3S1P51	—	—	—

3 – Results and Discussion

In silico Design of the Modular Toolkit Library

Level –1 acceptor vectors

For long term storage and housekeeping of promoter sequences, Level –1 acceptor vectors based on pSB1K5 were designed for subcloning linear promoter DNA fragments into a circular vector (Fig. 7). Using this plasmid, assembly of a Level –1 promoter part can be done using CPEC. These vectors are not necessary for the function of this toolkit, and linear DNA can be used directly as Level –1 parts. The end-user can opt to use these plasmids if they want to clone and upkeep in-house DNA stocks.

Level 0 acceptor vectors

The PL/PR assembly framework requires two Level 0 acceptor vectors: one for the PL position (5'-GGAG / ATCG-3' fusion sites) and one for the PR position (5'-ATCG / TACT-3' fusion sites). Both were designed based on the pSB1C5 backbone (high-copy pUC ORI, *CmR*), into which a GGA cloning site is inserted. This cloning site contains both *BsaI* sites required for Level 1 assembly, as well as *BsmBI* sites (5'-ACAT / TTGT-3' fusion sites) flanking the insert region such that any Level –1 promoter fragment can be inserted without modification (Fig. 7).

The pSB1C5 backbone was chosen because the chloramphenicol resistance cassette does not overlap with those used in downstream Level 1 and Level 2 assemblies, which allows clean positive selection at following assembly stages. However, the *CmR* gene within this cassette contains a *BsmBI* restriction site which competes with the *BsmBI*-mediated assembly of Level 0 parts. Thus, previously designed primers were used to destroy this site by performing a C435G mutation during plasmid re-ligation *via* Golden Gate Assembly (iG24-030 and iG24-031; Table 1).

Level 1 test device plasmids

Level 1 test device plasmids were designed based on the pJUMP27-1A backbone (Psc101 ORI, KanR). Two distinct promoter pairs were used to drive OR-gated expression of sfGFP-LVA: (i) P_{tacl} and P_{BAD} , regulated respectively by LacI and AraC; and (ii) P_{PsiTac1} and P_{m} , regulated by PsiR and XylS respectively. These two pairs were chosen to allow construction of functionally

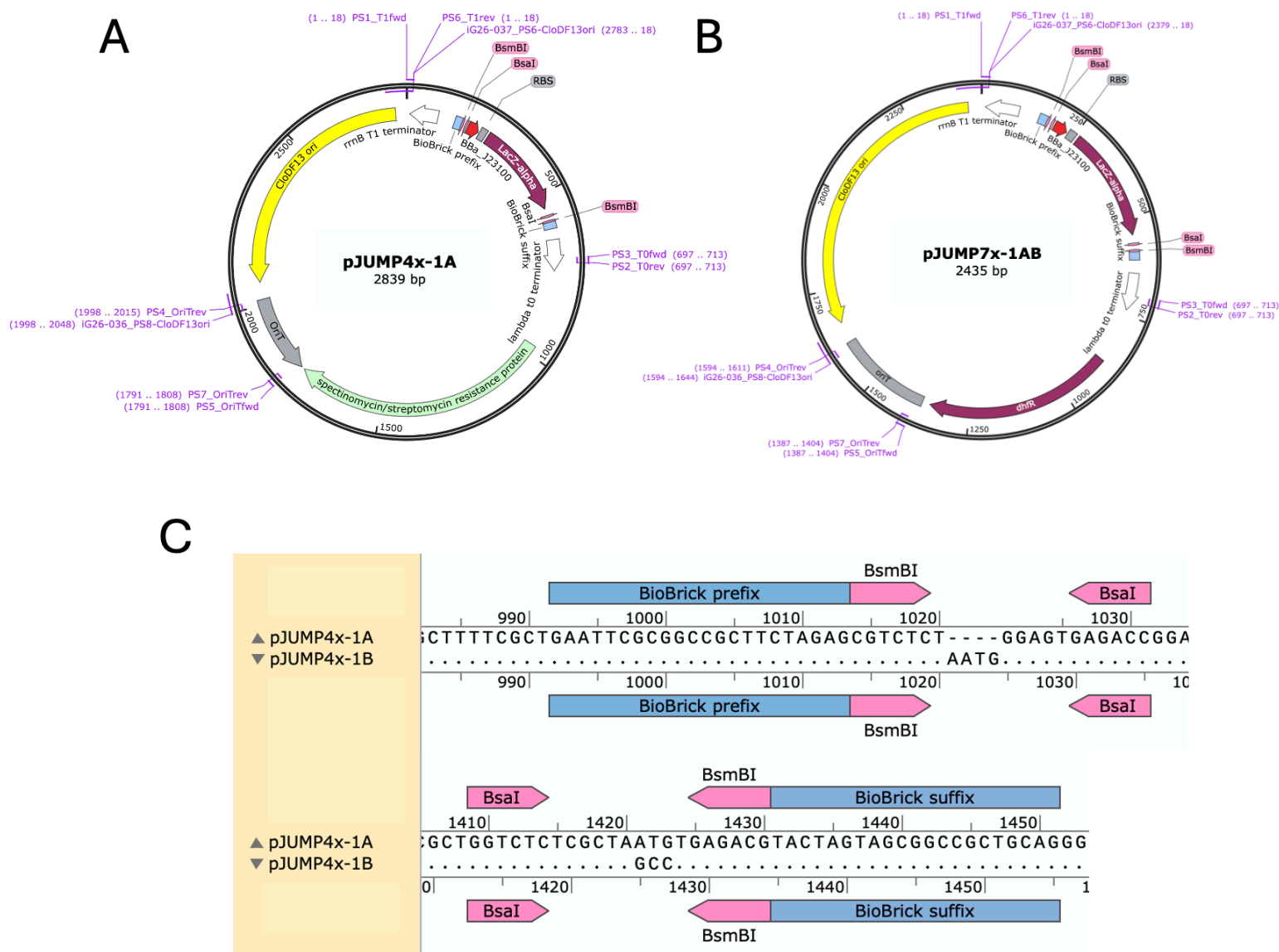


Figure 9. Representative plasmid maps for Level 1 and Level 2 accessory acceptor vectors. **A:** pJUMP4x-1A: Level 1 accessory AV. Contains CloDF13 ORI (yellow), LacZ- α screening marker (maroon), and SmR cassette (green). **B:** pJUMP7x-1AB: Level 2 accessory AV. Contains CloDF13 ORI (yellow), LacZ- α (maroon, top-right), and TmpR cassette (maroon, bottom). **C:** Alignment of pJUMP4x-1A and pJUMP4x-1B showing difference between up- and downstream GGA cloning site overhangs. *Visualized in SnapGene.*

equivalent circuits using orthogonal TF systems, aiding validation of the toolkit's general design. For each pair, PL and PR versions of each promoter were designed, allowing both single-promoter control constructs (one promoter driving sfGFP-LVA from either the canonical P position) and dual-promoter test constructs (one promoter in PL and the other in PR). This generated 8 OR-gate test plasmids with either single or dual-promoter arrangements for characterization of this system. AND-gated circuits are currently being designed and will be implemented in the near future by the hosting lab.

All test device reporter constructs employ sfGFP tagged with the LVA degradation peptide (sfGFP-LVA). The LVA motif confers instability upon the tagged protein, preventing it from accumulating during dynamic experiments and enabling kinetic monitoring of gene expression changes (Andersen et al., 1998).

Level 1 and 2 accessory vectors

LacI, AraC, PsiR, and XylS are not natively expressed at high levels in *E. coli* NEB 10-beta. Therefore, accessory expression plasmids encoding these TF genes were designed in addition to the test vectors. Compatible co-transformation of multiple plasmids within a single cell required careful selection of ORIs and antibiotic resistance markers.

As pJUMP27-1A carries the Psc101 ORI with KanR, the accessory plasmids were accordingly designed with the CloDF13 ORI (from MP6) and either a SmR cassette (from pSEVA481) or a TmpR cassette (from pSEVA721), enabling double antibiotic co-selection of both test and accessory plasmids (Fig. 9). The resulting series is designated pJUMP4x-1A through 1E (Level 1, single-TF constructs) and pJUMP7x-1AB through 1CD' (Level 2, dual-TF constructs); where "x" maps to the CloDF13 ORI, to follow the pJUMP naming convention (Valenzuela-Ortega and French, 2021). For this work, pJUMP4x-1A, pJUMP4x-1B, and pJUMP7x-1AB are the strictly required constructs, while the remaining designs were completed to provide flexibility for future characterization experiments.

For the purposes of this work, pJUMP4x plasmids are used as Level 1 acceptor vectors, and pJUMP7x as Level 2. pJUMP4x-1A and pJUMP4x-1B individual transcriptional units can be inserted into the pJUMP7x-1AB vector using BsmBI/Esp3I GGA due to their junction architecture. Six accessory plasmids were thus designed to supply the test device plasmids with the correct transcription factors required for logic gate functionality (Table 4). These

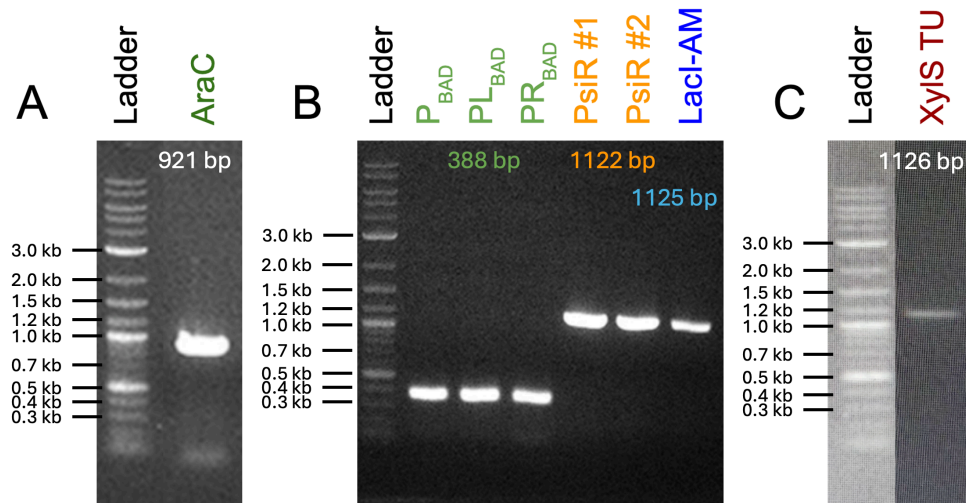


Figure 10. Gel electrophoresis of PCR amplicons. **A-C:** Bands representing successful amplification of target sequences. **A:** Lane 1: AraC (iG26-059 & iG26-060) on BBa_I0500. **B:** Lanes 1-3: P_{BAD} (iG26-053 & iG26-055), PL_{BAD} (iG26-053 & iG26-056), and PR_{BAD} (iG26-054 & iG26-055) on BBa_I0500. Lanes 4-5: PsiR (iG26-051 & iG26-052) on 2 different in-house plasmids. Lane 6: LacI-AM (iG26-057 & iG26-058) on BBa_K2675006. **C:** Lane 1: XylS(TU) (iG26-048 & iG26-049) on an in-house plasmid.

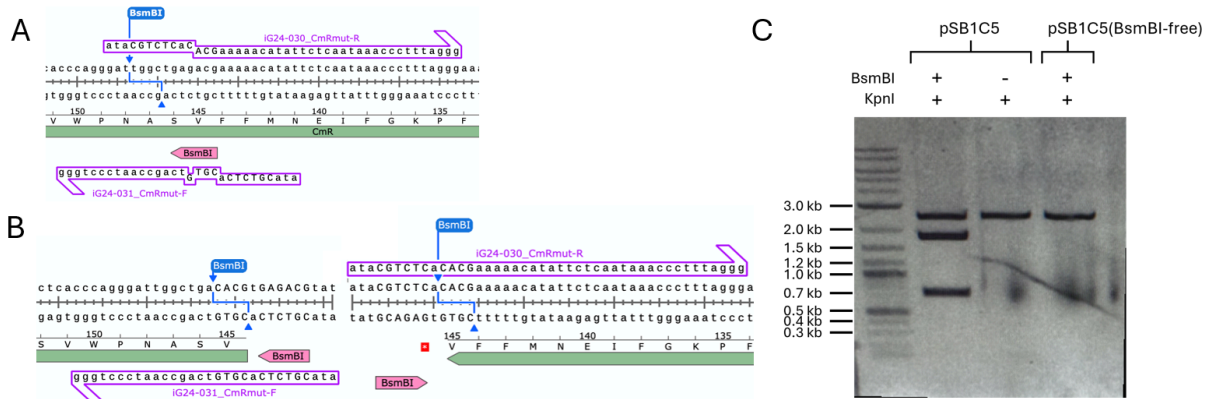


Figure 11. Mutation of illegal BsmBI site in pSB1C5 by GGA. **A:** Simulated hybridization of primers on pSB1C5. **B:** Simulated re-circularization of pSB1C5(BsmBI-free) amplicon using BsmBI. After digestion, CGTG overhangs hybridize and can be ligated. Within this overhang is the point mutation (C435G) that eliminates the BsmBI site. **C:** Gel electrophoresis of analytical digestion on pSB1C5(BsmBI-free). **Lanes 1-2:** pSB1C5, **Lane 3:** pSB1C5(BsmBI-free). There are 3 bands present in lane 1 due to incomplete digestion (single-cut) of the plasmid.

plasmids contain expression cassettes for each transcription factor individually, as well as dual-expression cassettes for test devices which utilize the PL/PR architecture.

Generation of DNA Fragments

All fragments required for Level 1 test device GGA reactions were generated prior to assembly. P_{BAD} was amplified in both PL and PR configurations from the iGEM part BBa_I0500 using primers appending the appropriate 4-nt GGA overhangs; AraC was amplified from the same template. sfGFP-LVA was amplified from an in-house plasmid. LacI-AM was amplified from iGEM part BBa_K2680537. PsiR amplification was performed from 2 different source plasmids due to plasmid labelling ambiguity. Gel electrophoresis confirmed that all amplified fragments produced bands of the expected sizes (Fig. 10). The promoters P_{tacI} , $P_{PsiTac1}$, and P_m , together with all synthetic RBS and terminator sequences, were ordered as linear DNA fragments from a specialized company and used directly in GGA reactions without prior cloning.

Construction of Backbone Vectors

BsmBI-free pSB1C5 acceptor vector

pSB1C5 sourced from in-house stocks was used as a template for primers designed by the 2024 iGEM Evry_Paris-Saclay team for site-directed mutagenesis and GGA cloning (iG24-030_CmRmut-R, iG24-031_CmRmut-F; Table 1) (Fig. 11A-B). Following purification of constructed plasmids, an analytical digestion using BsmBI and KpnI was performed to confirm the knockout of the BsmBI site. As can be seen in (Fig. 11C), the unmodified pSB1C5 plasmid is cut once by each BsmBI and KpnI. In a double digestion, this should lead to 2 bands, with the possibility of a third band for the incompletely digested (linearized) plasmid. This was observed for the double digestion condition, and the single digestion condition using only KpnI resulted in an identical linearized vector band. Double digestion of the mutated plasmid resulted in only one band, indicating that the BsmBI site has been removed successfully. Moreover, one clone was sent for whole-plasmid sequencing which revealed no errors in the construct. Sequencing confirmation of the C435G mutation is an important prerequisite for constructing a PL/PR Level 0 AV. Presence of a BsmBI restriction site within a Level 0 backbone would compete with the BsmBI sites used for Level -1 into Level 0 GGA reactions, and greatly reduce assembly efficiency. Therefore, using this backbone, I will be able to generate the library of genetic parts that can be distributed as part of the genetic toolkit.

CPEC assembly: initial attempt and troubleshooting

Construction of the pJUMP4x and pJUMP7x backbone plasmids was first attempted by multiple CPEC reactions between three linear fragments (AbR cassette, CloDF13 ORI, and JUMP GGA payload). However, during amplification of the JUMP payload region using primers PS1 and PS2, a non-specific band of ~1.1 kb was consistently obtained in gel electrophoresis (Fig. 12A). This size corresponds closely to the expected sfGFP-LVA size (1161 bp), which was indeed expected to be present in the JUMP vector templates from the iGEM kit. Sequencing of the resulting colony plasmids confirmed the absence of the expected JUMP payload architecture (Fig. 12B). This fact, taken with the absence of fluorescent colonies following culturing, suggests that the plasmids contained in the iGEM Distribution Kit did not contain an sfGFP CDS, leading to a shorter fragment than expected. Unfortunately, this highlights a risk of using DNA generated by external sources without prior sequencing for quality-control.

Following this, the JUMP payload fragment was synthesized as a DNA fragment encoding the required payload architecture and a LacZ- α blue/white screening marker in the interest of time. Three-fragment CPEC assembly was then repeated with the synthesized DNA. Colony PCR confirmed the correct insert size in the resulting constructs (Fig. 13). Moreover, as expected, positive colonies turned blue when cultured on media containing X-Gal. Whole-plasmid sequencing revealed the correct sequence identity of the constructed AV plasmids.

Construction of Accessory Expression Plasmids

Level 1 TF expression cassette assembly

Level 1 TF expression cassettes were assembled *via* BsaI GGA. Each cassette comprises a constitutive promoter, a synthetic RBS, a TF CDS (LacI-AM, AraC, XylS, or PsiR), and a synthetic terminator. Colony PCR (using PS1 and PS2 primers, Table 1) confirmed assembly of these constructs with successfully transformed colonies for each plasmid (Fig. 14). Whole-plasmid sequencing subsequently revealed 100% sequence identity for all constructs, with the exception of pL1Bs_LacI-AM_L3S1P51, which contained 3 missense mutations (M18Q, V43A, and Y161F). In fact, this reveals that this sequence is the wild-type *E. coli* (BBa_K091121) and not LacI-AM (BBa_K2680537) as we were informed. Further investigation is required, as this part also originates from the iGEM Distribution Kit. LacI-AM was chosen due to the presence of substitution mutations that increase its DNA binding affinity, thus effecting the

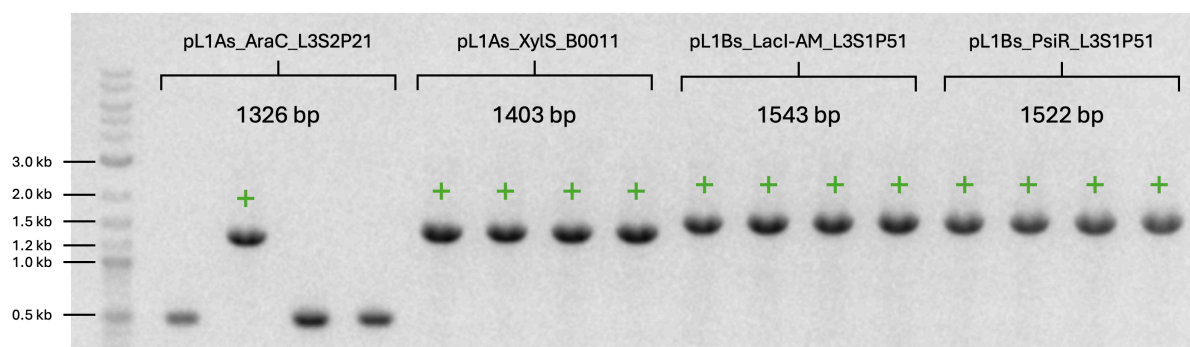


Figure 14. Colony PCR results for accessory plasmid GGA construction. Expected band sizes are annotated. Four colonies were analyzed for each construction, and bands marked with a green “+” represent positive colonies.

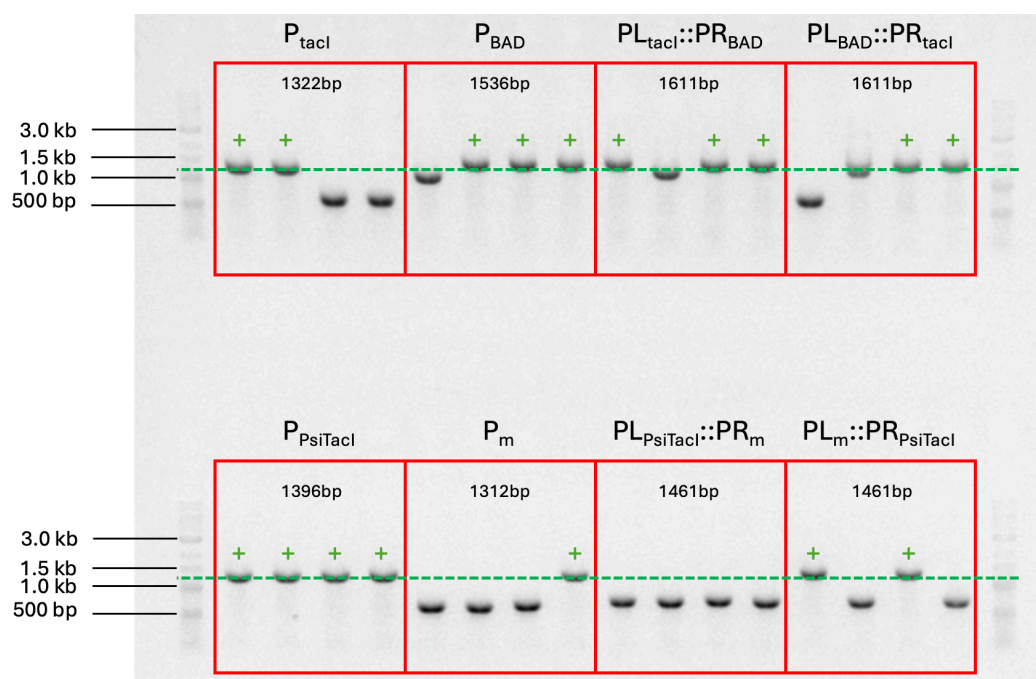


Figure 15. Colony PCR results for test plasmid GGA assemblies. Each construct contains the same RBS, sfGFP-LVA CDS, and terminator as detailed in Table 4. Expected band sizes are marked for each construct. Green dashed line can be used as a reference point to measure band size. Bands marked with a green “+” represent positive colonies.

IPTG dose-response dynamics for expression assay by reducing promoter leakiness. Thus, an intact LacI-AM sequence will need to be sourced to attempt this assembly again before continuing to characterization.

Level 2 dual-TF expression assembly

Two Level 2 dual-TF constructs are required for characterization experiments: pJUMP7x-1AB (LacI-AM + AraC) and pJUMP7x-1CD (PsiR + XylS). These are assembled by Esp3I GGA from the respective Level 1 TU inserts into the pJUMP7x backbone plasmid. At the time of writing, these assemblies have been performed, but the transformation has not been done. WPS must be done on these constructs before moving on to the characterization.

Construction of Level 1 Test Devices

All 8 Level 1 test device plasmids were assembled in parallel BsaI GGA reactions. Following overnight selection, colonies were obtained for all 8 assemblies. Colony PCR using junction-flanking primers confirmed the presence of correctly sized inserts in colonies from each assembly (Fig. 15). Following Miniprep purification of these plasmids, only four had sufficient concentration for whole-plasmid sequencing. Of these four, two were assembled correctly (pL1k_P_{BAD}_RiboJ-K2675017-T30_sfGFP_L3S2P56 and pL1k_P_m_RiboJ-K2675017-T30_sfGFP_L3S2P56). The third (pL1k_P_{PsiTacl}_RiboJ-K2675017-T30_sfGFP_L3S2P56) has a deletion in the RiboJ sequence (Δ AC), while the fourth (pL1k_P_{tacl}_P_{BAD}_RiboJ-K2675017-T30_sfGFP_L3S2P56) had the P_m sequence instead of PL_{tacl}/PR_{BAD} pair. This has likely arisen due to a pipetting error. Another round of WPS following a 100 mL culture plasmid purification of each of the test plasmids is in progress to confirm the assembly viability of the PL/PR architecture.

As this work serves as an early proof-of-concept for this assembly strategy, linear DNA fragments were used in place of plasmids wherever possible in the interest of minimizing the lead time before the test assemblies could be performed. Together, the successful construction and sequencing of all major components shows that the expanded MoClo framework works in practice and is compatible with standard assembly protocols. The major errors I encountered were both caught by sequencing and can be resolved. While confirmation of the PL/PR constructs specifically remains pending, the high sequence fidelity observed across all other assemblies is encouraging.

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4 – Conclusion and Perspectives

During this internship project, I produced a complete in silico design and near-complete construction of a modular genetic circuit toolkit for promoter circuit assembly in *E. coli*. I designed Level 0 acceptor vectors, Level 1 test devices, and Level 1/2 accessory expression plasmids in SnapGene, screened all parts for illegal restriction sites, verified GGA overhang compatibility, and chose the ATCG junction overhang for my novel PL/PR architecture based on published ligation fidelity data. Origin of replication and antibiotic resistance cassette combinations were also selected carefully to allow stable co-maintenance of multiple plasmids within the same cell. Constructions went largely to plan, with one exception, where the initial CPEC assembly of the accessory vector backbones failed because the iGEM kit vectors produced a spurious non-specific 1.1 kb band instead of the expected JUMP payload. Sequencing identified the issue, and switching to a synthesized DNA fragment resolved it. All 8 Level 1 test device plasmids were assembled by BsaI GGA and confirmed by colony PCR, as were the Level 1 accessory expression plasmids for LacI-AM, AraC, PsiR, and XylS. Once the pending Level 2 assembly and validation is complete, the toolkit will be ready for characterization.

Described in the 3rd objective of this project, the next step following sequencing confirmation of the genetic constructs is to establish dual-plasmid *E. coli* strains that carry both the test and accessory plasmids. After these strains are generated, their functional activity can be measured and characterized.

For characterization, *E. coli* cells harboring both a test device and an appropriate accessory vector will be grown overnight in 96-deep-well plates in LB medium with appropriate antibiotics, then induced during the exponential growth phase to ensure characterization captures gene expression dynamics under steady-state metabolism. Plates will be prepared using an OT-2 automated pipetting robot (Opentrons, New York) and incubated in a fluorescence- and OD-capable plate reader, with GFP fluorescence and OD₆₀₀ measured every 15 minutes. A fluorescence calibration assay will also be conducted to establish MEFL normalization curves using fluorescein standard solutions (Beal et al., 2020), ensuring measurements are expressed in absolute units and are comparable across instruments. Fluorescence values will be normalized by OD₆₀₀ and converted to MEFL/particle. To characterize the logic gates,

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dose-response assays will be performed by varying two effector concentrations independently across a 5×5 concentration matrix, yielding 25 conditions per circuit. With three biological replicates per condition, the remaining wells are reserved for negative controls, including cells harboring no circuit plasmid for auto-fluorescence correction.

Because I have one month remaining to complete the aforementioned work, obtaining characterization results is very feasible within the timeframe of this internship. The characterization of these circuits will establish a quantitative behavioral baseline for their further application in the host laboratory. The PL/PR promoter architecture represents a generalized extension to the MoClo standard that is not limited to the circuit topologies described in this work. In principle, any regulatory element that can be encoded within a promoter region can be adapted to this framework, opening new possibilities for the combinatorial design of synthetic gene regulation. The expanded-promoter MoClo toolkit directly addresses a gap in existing hierarchical assembly standards, where the promoter has until now been treated as an indivisible unit.

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Abstract / Résumé

Transcription factors are DNA-binding proteins that dynamically regulate gene expression. In prokaryotes, they control operons logically, lending them to applications in synthetic biology as programmable genetic logic gates. This internship developed a modular genetic circuit toolkit that extends the Modular Cloning (MoClo) Golden Gate Assembly standard. The key modification introduces a novel ATCG junction overhang between the canonical GGAG/TACT promoter overhangs of a Level 1 transcriptional unit, creating two independently exchangeable regions: the left promoter (PL) and the right promoter (PR). This architecture enables construction of AND, OR, and NOT logic gates, as well as dynamic circuits such as bistable switches. In this work, the toolkit was designed *in silico*, and all 8 Level 1 test device plasmids for the LacI/AraC and PsiR/XylS promoter systems were assembled by Golden Gate Assembly and confirmed by colony PCR. Accessory expression plasmids encoding the required transcription factors were constructed and sequence-verified at Level 1. Future perspectives include using the toolkit for dose-response characterization of assembled logic gates using sfGFP-LVA as a fluorescent reporter.

Keywords: *Gene regulation, genetic circuit, DNA toolkit, assembly standard, promoters.*

Acknowledgements

Thank you to Dr. Ioana Grigoras-Popescu for your mentorship and confidence. Thank you to the LiSSB at Genoscope for providing the consumables and instruments required by this work.

Thank you to my parents, for your love and support. I wouldn't have this opportunity without you.

Le chapitre suivant se lève.